# Policy Gradient Algorithms

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Suppose we have a policy parameterized by $\theta$, $\pi_\theta(\cdot|s)$

Example: Tabular policy: $\theta = \begin{pmatrix} \pi(a_1|s) \\ \vdots \\ \pi(a_k|s) \end{pmatrix}$ if there are $K$ actions.

Example: Neural network:

$$\pi(a|s) = \frac{e^{f_\theta(s,a)}}{\sum_{a'} e^{f_\theta(s,a')}},$$

where $f_\theta$ is a neural network.

Example: Log-linear policies: $f_\theta(s,a) = \phi^T(s,a)\,\theta$, where $\phi$ is known.

□

The goal is to solve

$$\max_\theta V_\theta(s_0), \qquad — Ⓐ$$

where $V_\theta$ is the value function associated with policy $\pi_\theta$ and $s_0$ is the initial state. We can solve the problem using gradient descent

$$\theta_{t+1} = \theta_t - \epsilon_t \nabla_\theta V_\theta(s_0)$$

if there is an expression for $\nabla_\theta V_\theta$ which can be computed from data. Note: it is difficult to provide convergence guarantees to the global optimum of Ⓐ since the problem lacks convexity or P-L except under some special conditions.

Policy gradient theorem: This theorem provides a nice expression for $\nabla_\theta V_\theta$ which allows its computation from data. Recall

$$V_\theta(s) = \sum_a \pi_\theta(a|s)\, Q_\theta(s,a)$$

$$\Rightarrow \boxed{\nabla_\theta V_\theta(s) = \sum_a \nabla_\theta \pi_\theta(a|s)\, Q_\theta(s,a) + \sum_a \pi_\theta(a|s)\, \nabla_\theta Q_\theta(s,a)}$$

Further, $Q_\theta(s,a) = r(s,a) + \alpha \sum_{s'} P(s'|s,a)\, V_\theta(s')$

$$\Rightarrow \boxed{\nabla_\theta Q_\theta(s,a) = \alpha \sum_{s'} P(s'|s,a)\, \nabla_\theta V_\theta(s')}$$

$\Rightarrow$

$$\nabla_\theta V_\theta(s) = \sum_a \nabla_\theta \pi_\theta(a|s)\, Q_\theta(s,a) + \alpha \sum_a \pi_\theta(a|s) \sum_{s'} P(s'|s,a)\, \nabla_\theta V_\theta(s')$$

$$= \sum_a \pi_\theta(a|s)\, \nabla_\theta \log \pi_\theta(a|s)\, Q_\theta(s,a) + \alpha \sum_a \pi_\theta(a|s) \sum_{s'} P(s'|s,a)\, \nabla_\theta V_\theta(s')$$

since $\nabla_\theta \log \pi_\theta(a|s) = \frac{1}{\pi_\theta(a|s)} \nabla_\theta \pi_\theta(a|s)$

$$= E_{\pi_\theta}\left(\nabla_\theta \log \pi_\theta(a|s)\, Q_\theta(s,a)\right) + \alpha\, E_{s'}\left(\nabla_\theta V_\theta(s') \,|\, s\right)$$

$\Rightarrow$

$$\nabla_\theta V_\theta(s) = E_{\pi_\theta}\left(\nabla_\theta \log \pi_\theta(a|s)\, Q_\theta(s,a)\right) + \alpha\, E_{s',\pi_\theta}\left(\nabla_\theta \log \pi_\theta(a'|s')\, Q_\theta(s',a') \,\middle|\, s\right) + \alpha^2 E_{s''}\left(\nabla_\theta V_\theta(s'') \,|\, s\right),$$

where $s''$ is the state two steps after starting in $s$.

Continuing:

$$\boxed{\nabla_\theta V_\theta(s) = E\left(\sum_{k=0}^{\infty} \alpha^k \nabla_\theta \log \pi(a_k|s_k)\, Q_\theta(s_k,a_k) \,\middle|\, s_0 = s\right)}$$

If $s_0 \sim \rho$, i.e., chosen according to some distribution $\rho$, and

$$J_\rho(\theta) \triangleq \sum_s V_\theta(s)\, \rho(s)$$

then

$$\boxed{\nabla_\theta J_\rho = E\left(\sum_{k=0}^{\infty} \alpha^k \nabla_\theta \log \pi(a_k|s_k)\, Q_\theta(s_k,a_k)\right)},$$

where the expectation now includes $\rho$.

Define $\nu_{\pi_\theta}(s,a) = (1-\alpha) \sum_{s_0} \rho(s_0) \sum_{k=0}^{\infty} \alpha^k P(s_k = s, a_k = a \,|\, s_0)$

Note that

$$\sum_{s,a} \nu_{\pi_\theta}(s,a) = (1-\alpha) \sum_{s_0} \rho(s_0) \sum_{k=0}^{\infty} \alpha^k \underbrace{\sum_{s,a} P(s_k = s, a_k = a \,|\, s_0)}_{=1}$$

$= 1$

$\nu_\pi$ is called the state-action visitation distribution. Hence, $\nabla_\theta J_\theta$ can also be written as

$$\boxed{\nabla_\theta J_\rho = \frac{1}{1-\alpha} \sum_{s,a} \nu_{\pi_\theta}(s,a)\, \nabla_\theta \log \pi_\theta(a|s)\, Q_\theta(s,a)}$$

Also, note that

$$\nu_{\pi_\theta}(s,a) = (1-\alpha) \sum_{s_0} \rho(s_0) \sum_{k=0}^{\infty} \alpha^k \sum_{s,a} P(s_k = s \,|\, s_0)\, \pi_\theta(a|s)$$

$$= \sum_a \pi_\theta(a|s) \underbrace{(1-\alpha) \sum_{s_0} \rho(s_0) \sum_{k=0}^{\infty} \alpha^k \sum_s P(s_k = s \,|\, s_0)}_{\triangleq\, \nu_{\pi_\theta}(s) \text{ (state visitation distribution; abusing notation)}}$$

$$= \sum_a \pi_\theta(a|s)\, \nu_{\pi_\theta}(s)$$

Further, note that

$$\sum_{s,a} \nu_{\pi_\theta}(s,a)\, \nabla_\theta \log \pi_\theta(a|s)\, V_\theta(s)$$

$$= \sum_s V_\theta(s) \sum_a \nu_{\pi_\theta}(s,a)\, \nabla_\theta \log \pi_\theta(a|s)$$

$$= \sum_s V_\theta(s)\, \nu_{\pi_\theta}(s) \sum_a \pi_\theta(a|s)\, \nabla_\theta \log \pi_\theta(a|s)$$

$$= \sum_s V_\theta(s)\, \nu_{\pi_\theta}(s) \underbrace{\sum_a \nabla_\theta \pi_\theta(a|s)}_{= \nabla_\theta \underbrace{\sum_a \pi_\theta(a|s)}_{=1} = 0}$$

$\Rightarrow$ $\nabla_\theta J_\rho$ can also be written as

$$\boxed{\nabla_\theta J_\rho = \frac{1}{1-\alpha} \sum_{s,a} \nu_{\pi_\theta}(s,a)\, \nabla_\theta \log \pi_\theta(a|s)\, A_\theta(s,a)},$$

where

$$A_\theta(s,a) = Q_\theta(s,a) - V_\theta(s),$$

i.e., the advantage of using action $a$ in state $s$ and using $\pi_\theta$ after that. This form is used in practice because when estimating the gradient from data, this form empirically shows lower variance.

Note: $A_\theta, Q_\theta, V_\theta$ are all short-hand notations for $A_{\pi_\theta}, Q_{\pi_\theta}, V_{\pi_\theta}$. I am using the short-hand notation for convenience. We could have used $\nu_\theta$ instead of $\nu_{\pi_\theta}$

Estimating $\nabla_\theta J_\rho$ from data:

For $k = 0, 1, 2, \dots, K-1$

Collect data using policy $\pi_{\theta_k}$ and initial distribution $\rho$, i.e., $(s_0^k, a_0^k, r_0^k), (s_1^k, a_1^k, r_1^k), \dots, (s_{T-1}^k, a_{T-1}^k, r_{T-1}^k)$

Estimate $V_{\pi_{\theta_k}}$ using TD-learning:

$$V_{\pi_{\theta_k}}(s) \approx f_w(s)$$

$w$ → parameters of a neural network or LFA.

$$\min \frac{1}{2} \Big(\underbrace{f_w(s_t^k) - r(s_t^k, a_t^k) - \alpha f_{w_t}(s_{t+1}^k)}_{\delta_t}\Big)^2$$

$$\Rightarrow w_{t+1} = w_t - \epsilon_t \delta_t \nabla_w f_w(s_t^k), \quad t = 0, \dots, T-1$$

set $w = w_T$

$$\widehat{\nabla_\theta J_\rho(\theta_k)} = \sum_{t=0}^{T-1} \alpha^t \nabla \log \pi_{\theta_k}(a_t|s_t)\, \hat{A}_t,$$

where $\hat{A}_t = \underbrace{r(s_t,a_t) + \alpha f_w(s_{t+1})}_{Q(s_t,a_t)} - \underbrace{f_w(s_t)}_{V(s_t)}$

$$\theta_{k+1} = \theta_k - \underset{\text{some step size}}{\tilde{\epsilon}_t}\, \widehat{\nabla_\theta J_\rho(\theta_k)} \qquad — Ⓘ$$

□

Clearly, there are many approximations involved here, such as the use of function approximation and the use of a finite trajectory of length $T$.

Note: In Ⓘ we need to make sure that $\pi_{\theta_{k+1}}$ is a valid policy. In other words, if $\pi_\theta$ is s.t. $\pi_\theta(a|s) \geq 0\ \forall a$ and $\sum_a \pi_\theta(a|s) = 1$ only for $\theta \in \Theta$ (some set), then one has to project $\theta_{k+1}$ into the set $\Theta$ whenever it lies outside the set.

Proximal Policy Optimization (PPO)

The most widely used policy gradient method is the PPO algorithm. To understand PPO (which is a heuristic), we will first provide an interpretation of gradient descent and prove something called the performance difference lemma. We will then be able to understand the rationale behind PPO.

Gradient Descent:

$$\min f(x)$$

Suppose we have an estimate $x_k$, to get the new estimate, we linearize $f$ around $x_k$ and add a "regularizer" to ensure that our new estimate is not too far from $x_k$.

i.e.,

$$\min_x f(x_k) + \nabla^T f(x_k)(x - x_k) + \frac{1}{2\epsilon_k}\|x - x_k\|^2$$

The solution is given by

$$\nabla^T f(x_k) + \frac{1}{\epsilon_k}(x_{k+1} - x_k) = 0$$

$$\Rightarrow \boxed{x_{k+1} = x_k - \epsilon_k \nabla f(x_k)}$$

Performance Difference Lemma

$$J_\rho(\theta) - J_\rho(\theta') = \frac{1}{(1-\alpha)} \sum_{s,a} \nu_{\pi_\theta}(s,a)\, A_{\pi_{\theta'}}(s,a)$$

Interpretation: $A_{\pi_{\theta'}}(s,a)$ is the advantage of using action $a$ in state $s$ over using $\pi_{\theta'}(\cdot|s)$. By averaging over $\nu_{\pi_\theta}(s,a)$, we are computing the advantage of using $\pi_\theta$ over $\pi_{\theta'}$.

Proof: $V_\theta(s) = E_{\pi_\theta}\left(\sum_{k=0}^{\infty} \alpha^k r(s_k,a_k) \,\middle|\, s_0 = s\right)$

$$V_{\theta'}(s) = \sum_{k=0}^{\infty}\left(\alpha^k V_{\theta'}(s_k) - \alpha^{k+1} V_{\theta'}(s_{k+1}) \,\middle|\, s_0 = s\right)$$

$$\Rightarrow V_\theta(s) - V_{\theta'}(s) = E_{\pi_\theta}\Big(\sum_{k=0}^{\infty} \alpha^k \underbrace{\big(\underbrace{r(s_k,a_k) + \alpha V_{\theta'}(s_{k+1})}_{Q_{\theta'}(s_k,a_k)} - V_{\theta'}(s_k)\big)}_{A_{\theta'}(s_k,a_k)} \,\Big|\, s_0 = s\Big)$$

$$= E_{\pi_\theta}\left(\sum_{k=0}^{\infty} \alpha^k A_{\theta'}(s_k,a_k) \,\middle|\, s_0 = s\right)$$

$$\Rightarrow \boxed{J_\rho(\theta) - J_\rho(\theta') = E_{\pi_\theta,\rho}\left(\sum_{k=0}^{\infty} \alpha^k A_{\theta'}(s_k,a_k)\right)}$$

□

Now, suppose we have an estimate of the optimal policy call it $\pi_{\theta_k}$. How do we obtain $\pi_{\theta_{k+1}}$?

PPO starts with performance difference lemma.

$$J_\rho(\theta_{k+1}) - J_\rho(\theta_k) = \frac{1}{(1-\alpha)} \sum_{s,a} \nu_{\pi_{\theta_{k+1}}}(s,a)\, A_{\pi_{\theta_k}}(s,a)$$

The goal is choose $\theta_{k+1}$ to maximize the RHS to increase the total discounted reward as much as possible. However, we cannot estimate the RHS from data because we need data from $\pi_{\theta_{k+1}}$, which don't have yet. So you approximate

$$\nu_{\pi_{\theta_{k+1}}}(s,a) = \nu_{\theta_{k+1}}(s)\, \pi_{\theta_{k+1}}(a|s)$$

$$\approx \nu_{\theta_k}(s)\, \pi_{\theta_{k+1}}(a|s),$$

so it approximates

$$J_\rho(\theta) - J_\rho(\theta_k) \approx E_{\substack{s \sim \nu_{\theta_k} \\ a \sim \pi_\theta}}\left[A_{\theta_k}(s,a)\right]$$

$$= E_{\substack{s \sim \nu_{\theta_k} \\ a \sim \pi_{\theta_k}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\theta_k}(a|s)} A_{\theta_k}(s,a)\right]$$

assuming $\pi_\theta$ and $\pi_{\theta_k}$ are close. To ensure this, PPO defines

$$\alpha_\theta(s,a) = \frac{\pi_\theta(s,a)}{\pi_{\theta_k}(a|s)}$$

and

$$\text{clip}(x) = \begin{cases} 1-\epsilon & \text{if } x \leq 1-\epsilon \\ x & \text{if } 1-\epsilon < x < 1+\epsilon \\ 1+\epsilon & \text{if } x \geq 1+\epsilon \end{cases}$$

and chooses $\theta_{k+1}$ as the solution to

$$\max_\theta E_{s,a \sim \nu_{\theta_k}}\left[\min\left(\alpha_\theta(s,a) A_{\theta_k}(s,a),\ \text{clip}(\alpha_\theta(s,a)) A_{\theta_k}(s,a)\right)\right]$$

The clip is sort of like regularization in gradient descent. clip + min above accomplish the following goals:

If $A_{\theta_k}(s,a) > 0$, then we want to increase the probability of such actions, but we don't want $\alpha_\theta$ to become too large, so $\alpha_\theta(s,a)$ is clipped to be $1+\epsilon$. If $-_{\theta_k}(s,a)$, the probability of such actions sho be decreased, but not much to ensure $\pi_{\theta_{k+1}} \approx \pi_{\theta_k}$. Hence, clip $\alpha_\theta$ to be $\geq 1-\epsilon$.